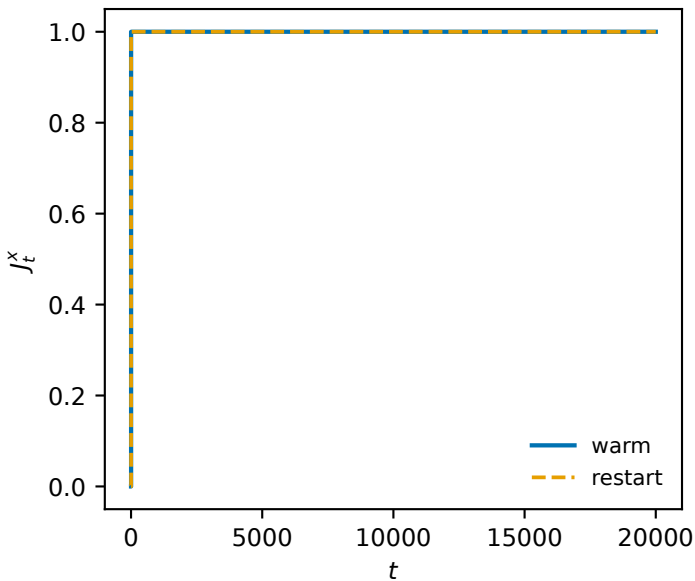
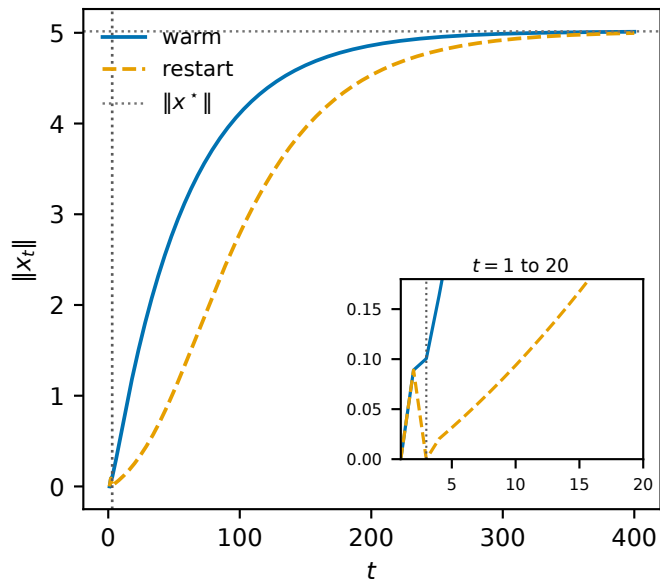


vs $y \equiv e_1$: finite doublingre-climb toward x^* regret at x^* : constant offset